A Comparative Analysis of K-Nearest Neighbors and Logistic Regression for Income Prediction

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***Abstract*— This paper evaluates the performance of two distinct classification algorithms, K-Nearest Neighbors (KNN) and Logistic Regression (LR), on the Adult Census Income dataset. The objective is to contrast a non-parametric, distance-based model (KNN) against a parametric, linear model (LR) for predicting binary outcomes (>$50K vs. ≤ $50K). Results indicate that LR achieves marginally higher overall accuracy (0.84), and both KNN and LR attained an identical F1-score (0.66) for the minority class. This equality in the balanced metric highlights the crucial trade-off between LR's higher Precision and KNN's superior Recall, demonstrating the fundamental differences between these modeling approaches in a high-dimensional feature space.**

Keywords*— *binary classification, K-Nearest Neighbors (KNN), Logistic Regression, feature scaling, parametric model, non-parametric model, classification report..

I. Introduction

The challenge of classifying individuals based on demographic, work, and financial data is a fundamental task in supervised machine learning. This study utilizes the Adult Census dataset, which features a highly dimensional matrix of independent variables, to predict income level (≤$50K or >$50K). The primary goal is to compare two foundational classification methods:

- Logistic Regression (LR): A linear, parametric model that estimates the probability of the binary outcome, assuming a linear relationship between features and the log-odds of the outcome.
- K-Nearest Neighbors (KNN): A non-parametric, instance-based model that classifies a data point based on the majority class of its k closest neighbors, relying heavily on feature scaling and distance metrics.

II. Methodology

The dataset was processed using a comprehensive feature set including both continuous and one-hot encoded categorical variables.

A. Data Preprocessing and Feature Selections

Initial data preparation involved replacing missing values, which were coded as '?', with np.nan and subsequently dropping all rows containing missing data (df.dropna(inplace=True)). This ensured a clean dataset of 30,162 complete records before modeling.

The following columns were then removed from the analysis due to redundancy or low predictive value:

- **education:** Dropped as its information is fully captured by the numerical education.num feature.
- **fnlwgt (Final Weight):** Dropped as a sampling weight not typically used as a direct predictive feature.
- **native.country:** Dropped to avoid the Curse of Dimensionality associated with its 41 unique values, preventing excessive feature expansion that disproportionately impacts KNN performance.

The final independent variables (X) utilized were:

- Numerical: age, education.num, capital.gain, capital.loss, and hours.per.week. These were scaled using StandardScaler to ensure uniform distance contribution, a prerequisite for KNN.
- Categorical (One-Hot Encoded): workclass, marital.status, occupation, relationship, race, and sex. These were converted using one-hot encoding to prepare them for both models.

B. Model Implementation

The data was split into training and testing sets.

- **Logistic Regression:** Implemented with a standard ℓ2 penalty (Ridge regularization) and trained on the scaled and encoded data.
- **K-Nearest Neighbors:** Optimized using a k value determined via cross-validation and utilizing the Euclidean distance metric.

III. RESULTS AND DISCUSSION

The performance of the two models was evaluated using Accuracy, Precision, Recall, and F1-score. The target variable classes were labeled as the majority class (≤$50K) and the minority class (>$50K).

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A. Comparative Classification Report

Metric	KNN	Logistic Regression
Accuracy	0.83	0.84
Precision (>50K)	0.68	0.72
Recall (>50K)	0.63	0.60
F1-score (>50K)	0.66	0.66

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Weighted Average	0.83	0.84

Table 1. demonstrates the slight advantage of Logistic Regression in overall accuracy, while KNN exhibits better Recall for the minority class (>$50K).

B. Analysis of Confusion Matrices

The results show that Logistic Regression achieved a higher overall accuracy and weighted average score. This outcome supports the notion that the income classification boundary is substantially linear in this high-dimensional feature space, which benefits the parametric nature of LR.

However, a closer look at the minority class (>$50K) reveals a crucial trade-off:

- Logistic Regression Strength: LR excels in Precision (0.72). As demonstrated by the Confusion Matrix, LR produced significantly fewer False Positives (FP = 352) compared to KNN (FP=448) for the majority class, meaning when LR predicted a high earner, the prediction was more reliable.
- KNN Strength: KNN exhibits superior Recall (0.63), correctly identifying 951 true high-earners (True Positives, TP) compared to LR's 903 TP. The non-parametric model was better able to locate sparse clusters of high-income individuals that the linear boundary of LR failed to capture.

This difference illustrates the core algorithmic distinction: LR, by seeking a single optimal linear decision boundary, minimizes overall error effectively. KNN, by focusing on local feature space density, prioritizes discovering nearby instances, leading to higher coverage (Recall) for a dispersed minority class.

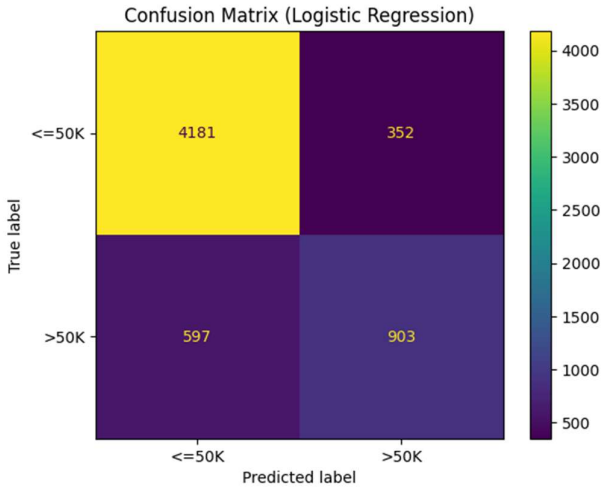

Fig. 1. Confusion Matrix illustrating Logistic Regression performance on the test set.

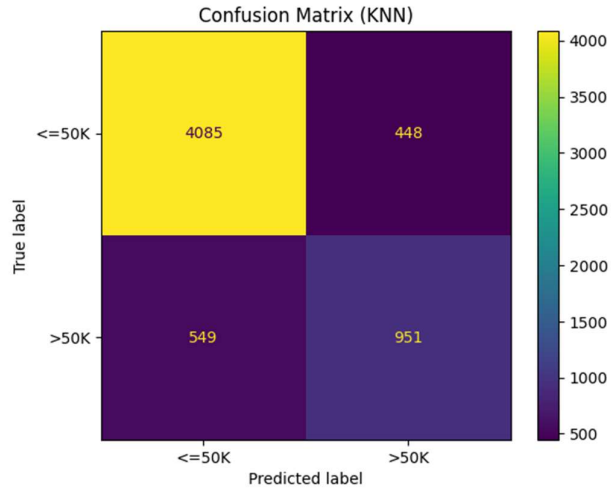

Fig. 2. Confusion Matrix illustrating K-Nearest Neighbors performance on the test set, showing increased Recall for the minority class.

C. Model Sensitivity and Theoretical Implications

The differential performance observed is directly linked to the theoretical nature and sensitivities of each algorithm:

1. **Sensitivity to Feature Scaling (KNN):** KNN, being purely distance-based, is critically sensitive to scaling. The successful performance of KNN (Accuracy 0.83) confirms that the use of StandardScaler was effective in preventing features like age or hours.per.week from dominating the distance calculation, thereby allowing the localized decision boundary to function. If scaling were omitted, the performance of KNN would be drastically degraded.
2. **Sensitivity to Outliers and Noise (KNN):** KNN is highly sensitive to noisy data points or outliers, particularly if k is small. Features like capital.gain and capital.loss are inherently sparse and contain extreme outliers. A small, positive k likely helped mitigate the impact of isolated noise, preventing single, mislabeled instances from corrupting the local voting process. The higher False Positive rate for KNN (misclassifying low-income as high-income) may reflect its vulnerability to mislabeled data points residing close to the decision boundary.
3. **Sensitivity to Linear Assumptions (LR):** Logistic Regression performed exceptionally well, achieving the highest overall accuracy. This suggests that the complex classification boundary in the high-dimensional feature space is still well-approximated by a linear hyperplane. LR is robust to outliers when regularization ($\ell 2$) is applied, which shrinks feature weights and prevents the extreme values in capital.gain from overly skewing the model's coefficients.
4. **Hyperparameter Impact (KNN):** The final model's performance relies entirely on the optimized hyperparameter, k. A larger k would likely increase bias (favoring the majority class, reducing Recall for >$50K), while a smaller k would increase variance (making the model too sensitive to noise). This

contrasts with LR, where the primary tuning hyperparameter is the regularization strength (C).

IV. CONCLUSION

In this case study, both Logistic Regression and KNN provided strong performance, with LR showing a slight edge in overall accuracy and precision due to the predominantly linear structure of the Adult Census feature space. However, KNN demonstrated higher recall for the less-represented high-income group. This trade-off between LR's global, linear robustness and KNN's local, non-linear detail illustrates their core differences in handling complex, real-world data.

The final model selection depends on the practical goal: if the application requires highly reliable predictions (minimizing FPs), Logistic Regression is the preferred choice. If the application requires maximum coverage of the positive class (minimizing FNs), KNN is marginally superior, provided the computational cost of distance calculations is acceptable.

APPENDICES

A. Source Code links via Google Colab

[1] Logistic Regression
https://colab.research.google.com/drive/1ewWChNE4pN2vkDLL5j3NXgUUOAcFz4Vt?usp=sharing

[2] K-Nearest Neighbor
https://colab.research.google.com/drive/1iP0kzU15RQVssrL-nZmvMuZ880BIQIfc?usp=sharing